

Autonomous Credit Intelligence: An Event-Driven, Self-Healing Multi-Agent Framework for Real-Time B2B Risk Assessment

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Abstract

Intraday B2B credit monitoring requires continuous synthesis of payment behaviour, balance-sheet fundamentals, and litigation signals. Given the proprietary confidentiality of commercial enterprise trade ledgers, evaluating streaming credit architectures necessitates reproducible macroeconomic stress simulation. This paper presents ACIS-X, a reactive event-driven multi-agent framework on Apache Kafka that continuously maintains a live per-customer risk score. Specialized agents coordinate over partitioned Kafka topics across five core layers, an autonomous model governance plane, and a self-healing plane. ACIS-X combines Kafka Exactly-Once Semantics (EOS) with idempotent database sinks, three-tier signal fusion, SHAP audit logging, streaming drift tracking via Population Stability Index (PSI), and heartbeat-driven recovery from committed offsets. Evaluated under a jump-diffusion stress harness, ACIS-X demonstrates sub-50 ms streaming latency (P95 16.2 ms), near-linear consumer scaling, and autonomous fault recovery with storm prevention.

Keywords: Multi-agent systems, Event-driven architecture, Credit risk intelligence, Apache Kafka, Exactly-Once Semantics, Self-healing systems

1 Introduction

Enterprise credit risk monitoring demands continuous evaluation of payment dynamics, financial health, and litigation. Event streaming platforms like Apache Kafka [1] enable real-time risk updates, but recent LLM-based multi-agent systems [2] introduce multi-second latencies (> 2 s) and non-deterministic outputs unsuitable for sub-50 ms financial streaming. Conventional Kafka pipelines perform static transaction scoring without agentic autonomy or drift governance.

A fundamental challenge is that real B2B invoice ledgers and defaults are confidential commercial secrets. We therefore separate *architectural validation* (the core contribution) from *empirical default forecasting*. We introduce ACIS-X (Autonomous Credit Intelligence System, Extended), an event-driven multi-agent framework grounded in the Actor model, with the following contributions:

- An event-driven multi-agent architecture on Kafka with autonomous self-healing and model governance.
- Kafka stream-level EOS combined with idempotent persistence and action deduplication for external sinks.
- Three-tier signal fusion integrating behavioural metrics, financial ratios, and litigation with deterministic risk policy guardrails.
- Streaming model governance via PSI drift tracking and online challenger benchmarking, inspired by SR 11-7.
- Systems benchmarks measuring throughput, scaling, P95/P99 latency, and fault recovery under a jump-diffusion stress harness.

2 Related Work and Research Gap

Corporate credit scoring—from Z-scores [3] to explainable ML [4, 5]—relies on offline batch feature stores incapable of capturing intraday delinquency transitions. Kafka event sourcing is adopted for sub-second fraud scoring [6, 7], yet existing architectures evaluate isolated transactions rather than stateful, multi-source credit profiles combining balance-sheet ratios and litigation.

Cognitive MAS frameworks—MASCA [2], Kubam’s credit agent [8], Kou et al. [9]—applied LLMs and MARL to credit assessment, but their multi-second execution times and prompt non-determinism are incompatible with high-throughput, low-latency streaming. Existing autonomic resilience [10, 11] recovers at coarse container boundaries rather than providing per-agent heartbeat recovery with offset replay. Table 1 positions ACIS-X at the intersection of streaming fusion, deterministic compliance, autonomous governance, and fine-grained agent recovery.

Table 1 Architectural positioning across credit risk paradigms.

Dimension	Batch XAI [4]	Stream [7]	Cog. MAS [2]	ACIS-X
Latency	Hours-Days	<1 s	>2 s	P95 16 ms
Signal Scope	Fin. ratios	Single txn	Text/reports	3-tier fusion
Exec. Model	Batch	Stream proc.	LLM/MARL	Reactive MAS
Atomicity	DB ACID	AL-once/EOS	None	EOS + PK
Fault Recov.	Cold restart	Node failover	Prompt retry	HB + offset
Drift Gov.	Offline	Ext. dash.	Manual	PSI + Challenger

3 System Architecture

ACIS-X maintains a live risk score for each customer c_i :

$$\mathcal{R}(c_i, t) = \mathcal{F}(\mathcal{B}(c_i, t), \mathcal{E}(c_i, t), \mathcal{L}(c_i, t)) \in [0, 1], \quad (1)$$

where $\mathcal{B}$ is behavioural history, $\mathcal{E}$ external financial indicators, and $\mathcal{L}$ litigation exposure. Design targets include sub-second processing, usable scores under partial signals, zero silent loss of committed work, and idempotent duplicate handling.

The architecture comprises six layers on a common Kafka event bus: Layer 1 ingests customers, invoices, payments, and behavioural indicators; Layer 2 enriches via financial APIs and litigation scraping; Layer 3 performs payment risk prediction and composite scoring; Layer 4 enforces credit policy and collections; Layer 5 persists state; Layer 6 tracks drift and benchmarks challengers. A system plane manages time ticks, monitoring, self-healing, and dead-letter-queue isolation. All events are keyed by `customer_id`, ensuring per-customer partition affinity and temporal ordering without distributed join coordinators.

Table 2 catalogues every agent, its layer, and operational role. Each operates as an isolated OS process with a dedicated consumer group.

4 Methodology

4.1 Three-Tier Signal Fusion

Tier 1 – Behavioural signals. The customer state agent rebuilds metrics from invoices and payments under a per-customer lock. With payment/due timestamps $\text{paid}_k, \text{due}_k$:

$$\begin{aligned} \text{OTR}(c_i) &= \frac{|\{I_k : \text{paid}(I_k) \leq \text{due}(I_k)\}|}{|\mathcal{I}(c_i)|}, \\ \text{AvgDelay}(c_i) &= \frac{1}{|\mathcal{P}|} \sum_k \max((\text{paid}_k - \text{due}_k), 0). \end{aligned} \quad (2)$$

Tier 2 – External signals. Financial ratios are cached for 90 days as $\mathcal{E}(c_i) \in [0, 1]$; litigation scraping scores court exposure over 365 days as $\mathcal{L}(c_i) \in [0, 1]$.

Table 2 ACIS-X agent inventory: layer and operational role.

Agent	L	Role
ScenarioGenerator	1	Emits reproducible synthetic/jump-diffusion macro-stress streams.
CustomerState	1	Rebuilds behavioural metrics (OTR, delay, utilisation).
ExternalData	2	Enriches and caches financial ratios (D/E, ROE, PE).
LitigationScraping	2	Quantifies legal exposure from court/news feeds.
Aggregator	2	Fuses Tier 2 financial and litigation signals (0.6/0.4).
PaymentPrediction	3	RF scoring + guardrails + SHAP attributions.
RiskScoring	3	Refines base score with behavioural & temporal velocity.
OverdueDetection	4	Time-tick driven delinquency detection.
CreditPolicy	4	Maps risk tiers to operational intents.
Collections	4	Idempotent collection actions; storm prevention.
DBAgent	5	Persists events (SQLite/Postgres) for recovery.
Query / Memory	5	In-memory reads with DB fallback.
ModelGovernance	6	PSI/KS drift tracking; challenger benchmarking.

Tier 3 – Fusion. The aggregator fuses external signals:

$$R_{\text{ext}}(c_i) = w_{\text{fin}} \cdot \mathcal{E}(c_i) + w_{\text{lit}} \cdot \mathcal{L}(c_i) \in [0, 1], \quad (3)$$

with $w_{\text{fin}}=0.60$, $w_{\text{lit}}=0.40$. Sensitivity is evaluated in Section 6.2.

4.2 Payment Risk Prediction and Guardrails

The feature vector $X \in \mathbb{R}^d$ includes normalised delay, credit utilisation, invoice ratio, late rate, external financial score, litigation index, and rating penalty. A Random Forest ($M=50$ trees), pre-calibrated offline, yields $r_{ml} = \frac{1}{M} \sum_m h_m(X)$. ACIS-X operates **label-free at runtime**:

$$r_{\text{final}} = \max(0, \min(1, r_{ml} + r_{\text{policy}})), \quad (4)$$

where deterministic guardrail r_{policy} enforces rating penalties (+0.10 B, +0.20 C) and litigation caps (+0.15 if $\mathcal{L}>0.1$, +0.25 if $\mathcal{L}>0.5$). Attributions are logged via SHAP TreeExplainer [12] for auditability.

4.3 Risk Refinement and Self-Healing

The risk scoring agent forms adjusted risk from prediction r_{pred} :

$$r_{\text{adj}} = \max\left(0, \min\left(1, r_{\text{pred}} + 0.85 \cdot \Delta_{\text{beh}} + 0.30 \cdot \Delta_{\text{temp}}\right)\right), \quad (5)$$

where $\Delta_{\text{beh}} = 0.40a_{\text{overdue}} + 0.20a_{\text{pay}} + 0.15a_{\text{delay}} + 0.15a_{\text{out}} + 0.10a_{\text{exp}}$ ranks urgency across normalised factors, and $\Delta_{\text{temp}} \in [-0.2, 0.3]$ captures trend direction. Agents emit heartbeats every $T_H=10$ s; monitoring detects a 60 s timeout (6 missed beats), triggering restart from the last committed offset.

4.4 Model Governance and Stress Harness

Layer 6 tracks predictions across sliding windows ($W=200$) and computes PSI:

$$\text{PSI} = \sum_{b=1}^B (P_b - Q_b) \cdot \ln\left(\frac{P_b}{Q_b}\right), \quad (6)$$

where P_b, Q_b are baseline/current quantile proportions ($B=10$ bins). Values $\text{PSI} \geq 0.10$ trigger drift warnings. A challenger model (linear SGD) is continuously benchmarked against the champion RF.

To evaluate without confidential ledgers, a jump-diffusion stress harness generates scenarios:

$$dZ_t = \kappa(\mu - Z_t) dt + \sigma_Z dW_t + J_t dN_t, \quad (7)$$

where $Z_t \in [0, 1]$ is a latent stress factor, κ the mean-reversion rate, σ_Z volatility, and $J_t dN_t$ a Poisson jump (liquidity shocks). Customer delinquency is conditioned on Z_t via Pareto delay mixtures across Calm, Stressed, and Crisis regimes.

5 Implementation

ACIS-X is implemented in Python 3.11 (~55 files, seven packages). Agents launch as separate OS processes managed by supervisor (max 5 attempts / 2 min). Messages use Pydantic-validated schemas (v1.1) with correlation IDs; validation failures route to a dead-letter queue. Kafka Stream EOS (transactional producers, `read_committed` consumers) provides exactly-once for consume-transform-produce loops. External sinks use composite-key upserts (ON CONFLICT DO NOTHING) and LRU deduplication caches.

6 Experimental Design and Evaluation

6.1 Systems Metrics

Tests executed on Windows 11 (i7-12700H, 32 GB RAM, Docker Kafka 3.6): **180/180 tests passed** (100%). Table 3 reports system metrics under streaming load across 6 partitions.

6.2 Ablation and Sensitivity

Ablation across $n=200$ customers over 42 seeds (Table 4): behavioural refinement provides the dominant lift ($\Delta\rho = +0.065$, paired $t=37.29$, $p<0.0001$, $df=41$), while external enrichment contributes marginally ($\Delta\rho = +0.005$). In real B2B credit, external

Table 3 System performance, latency, and scalability.

Metric	Value	Scope
Peak Throughput	6,420 evt/s	6 partitions, batched
Latency (P95/P99)	16.2 ms / 28.4 ms	Full 5-layer pipeline
Consumer Lag	0 (steady); <15 ms post-burst	Recovery
Dedup (10k repeats)	1 action dispatched	Idempotency
Scaling ($N=4/6$)	$3.49\times/4.93\times$	Partition-bounded
Governance PSI	4.1 ms (<0.10 stable)	Sliding-window
Suite Pass Rate	180/180 (100%)	Full verification

financials update quarterly and serve as low-frequency policy guardrails. SHAP adds ~ 0.9 ms/event.

Table 4 Architectural ablation ($n=200$, 42 seeds): Spearman ρ with ground truth.

Configuration	ρ	$\Delta\rho$	Note
Full ACIS-X	0.734 ± 0.035	—	All layers
–External enrich.	0.728 ± 0.035	–0.005	$p<0.0001$
–Behav. refine.	0.669 ± 0.042	–0.065	$p<0.0001$
–SHAP audit	(latency)	+0.9 ms	Overhead

Varying fusion weights ($w_{\text{fin}}/w_{\text{lit}}$: 50/50, 60/40, 70/30) yielded $\rho=0.760$, 0.872, 0.941 respectively, confirming the pipeline responds predictably to policy weight shifts without online RL.

6.3 Fault Injection

Table 5 reports self-healing evaluation. Internal decision latency is sub-millisecond; cooldown prevents restart storms (exactly 1 restart/agent in 5-agent cascades). In distributed deployment, total recovery is governed by the 60 s heartbeat timeout and Kafka rebalance (3–5 s), after which consumers resume from committed offsets.

Table 5 Fault injection results (single-node decision latency).

Scenario	Metric	Value	Assert.
Agent crash	decision lat.	0.26 ms	<100 ms
Latency spike (5 s)	decision lat.	0.10 ms	<100 ms
5-agent cascade	restarts/agent	1	exactly 1
100-cycle endurance	success rate	100%	100/100
100-cycle endurance	P95 lat.	0.23 ms	<50 ms

6.4 Predictive Comparison

Table 6 benchmarks ACIS-X against supervised learners ($n=500$, 42 seeds). Supervised models reach high AUC (>0.96) by training on synthetic generator labels unavailable in production. ACIS-X operates label-free, improving upon heuristic rules ($F_1=0.757$ vs. 0.738, winning 32/42 seeds) while providing continuous ranking (AUC = 0.861) at sub-50 ms latency.

Table 6 Predictive benchmark ($n=500$, 42 seeds). Supervised models use ground-truth labels; ACIS-X is label-free.

Method	F1	AUC
Naive utilisation	0.738 ± 0.038	—
ACIS-X streaming (label-free)	0.757 ± 0.035	0.861 ± 0.032
Log. Reg. (basic/enriched, sup.)	0.927/0.965	0.988/0.998
Random Forest (basic/enriched)	0.886/0.895	0.967/0.972
Gradient Boosting (basic/enriched)	0.879/0.888	0.967/0.970

7 Discussion

The evaluation validates ACIS-X as a low-latency, resilient multi-agent architecture for real-time credit intelligence. Behavioural transaction dynamics drive intraday differentiation ($\Delta\rho=+0.065$, $p<0.0001$), while external ratios act as low-frequency guardrails. PSI and SHAP (<1 ms) run inline without bottlenecking throughput.

7.1 Limitations

Domain Data: Public retail credit benchmarks lack B2B invoice dynamics; ACIS-X’s primary contribution is the situational streaming pipeline evaluated under jump-diffusion tail-risk scenarios. **Distributed Recovery:** The 0.26 ms metric reflects single-node decision latency; multi-node recovery is governed by 60 s heartbeat timeout and Kafka rebalances (3–5 s). **Governance:** Automated PSI/challenger signals require coupling with human-in-the-loop credit committee review.

8 Conclusion and Future Work

ACIS-X presents an event-driven multi-agent credit architecture on Kafka featuring three-tier signal fusion, stream-level EOS with idempotent sinks, autonomous model governance, SHAP audit persistence, and heartbeat-driven recovery. Across 42 seeds, the pipeline achieved 6,420 events/s peak throughput, P95 16.2 ms latency, and $4.93\times$ partition scaling. Behavioural refinement provides the primary ranking lift ($\Delta\rho=+0.065$) with <1 ms SHAP overhead, and fault injection confirms sub-millisecond self-healing decisions. Future work targets field validation via anonymized ERP partnerships and distributed disaster recovery drills.

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Declarations

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References

- [1] Kreps, J., Narkhede, N., Rao, J.: Kafka: A distributed messaging system for log processing. In: Proceedings of the NetDB Workshop (2011)
- [2] Jajoo, G., Pandey, A., Chitale, P., Agarwal, S.: MASCA: LLM-based multi-agent system for credit assessment. arXiv preprint arXiv:2507.22758 (2025)
- [3] Altman, E.I.: Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance* **23**(4), 589–609 (1968)
- [4] Chang, V., Xu, Q.A., Akinloye, S.H., Benson, V., Hall, K.: Prediction of bank credit worthiness through credit risk analysis: An explainable machine learning study. *Annals of Operations Research* **354**, 247–271 (2025)
- [5] Nwafor, C.N., Nwafor, O., Brahma, S.: Enhancing transparency and fairness in automated credit decisions: An explainable novel hybrid machine learning approach. *Scientific Reports* **14**, 25174 (2024)
- [6] Kesarpu, S., Dasari, H.P.: Kafka event sourcing for real-time risk analysis. *International Journal of Computational and Experimental Science and Engineering* **11**(3) (2025)
- [7] Parnerkar, H.: Comparative study of event-driven architectures in financial systems for real-time risk analysis and mitigation. *European Journal of Electrical Engineering and Computer Science* **9**(6), 46–54 (2025)
- [8] Kubam, A.: Agentic AI for autonomous, explainable, and real-time credit risk decision-making. arXiv preprint arXiv:2601.00818 (2026)
- [9] Kou, G., *et al.*: Learning to fuse: Cost-sensitive credit assessment via hierarchical multi-agent reinforcement learning. In: Proceedings of the NeurIPS GenAI in Finance Workshop (2025)
- [10] Beereddy, K.R.: Self-healing streaming architecture: Resilience patterns for event-driven microservices. *Journal of Computer Science and Technology Studies* **7**(8), 254–268 (2025)
- [11] Kephart, J.O., Chess, D.M.: The vision of autonomic computing. *Computer* **36**(1), 41–50 (2003)
- [12] Lundberg, S.M., Erion, G., Chen, H., *et al.*: From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence* **2**(1), 56–67 (2020)